SOUP INTERNSHIP · TAKE-HOME TASK

Frontend Engineer

Work: 4–6 h · Deadline: 48 h · Any stack · Deployment required (Vercel or GitHub Pages)

DATA

Generate it yourself: 20 training runs with the fields step, loss, lr, grad_norm. Some are deliberately broken: NaN and Infinity in loss, duplicate and unsorted steps, gaps, one run with 1,000,000 points and a single outlier spike in the middle of a long series.

TASK

A training-run monitor, and an answer to one question: can you show it to someone deciding whether to stop a $500 run? A chart that looks right and a chart that is truthful are not the same thing.

1. Build it. A run list (sort, filter), a run page with loss, lr and grad_norm charts, summary cards, loading, empty and error states, responsive layout, keyboard control, sufficient contrast.

2. Prove the chart is truthful. You can't draw a million points as they are, so you're discarding something. Show with automated tests that the spike is visible at every zoom level, NaN isn't drawn as 0 or silently smoothed, duplicates and unsorted steps don't distort the line, and numbers and axis labels match the data. Include the output and state what the tests would NOT catch.

3. Performance in numbers. First render and response to zoom and hover in a Performance trace (CPU throttling ×4), plus tab memory at 1M points. Numbers before and after optimization.

4. Silent failures. What looks fine but is wrong: stale data, a request race when switching runs quickly, locale and time zone, smoothing that hides divergence, a log scale with zeros. How you checked.

5. Verdict. SHIP or DON'T SHIP, and what to change first.

DELIVERABLES

Repository and deployed link · tests and their output · the Performance trace (file or screenshots) · a note of ≤2 pages · one paragraph on what surprised you and what still concerns you. Submit by email reply to team@trysoup.dev. Don't edit or hide failed results: a clear negative beats a quiet success. If something breaks, tell us what happened, what you tried, and what you learned.

AI TOOLS

Allowed (we use them too). Note where you used them, which suggestions you accepted, and what you verified, changed, or redid yourself. Not a penalty: we want to see how you work and where you spot an unreliable tool.

LIVE DISCUSSION

40 minutes, after submission. Don't pre-write answers; we want to think it through with you. Possible questions: The chart is smooth: how do you know it's honest? How would you draw 5M points and lose nothing? What do you show instead of NaN, and why? What do you keep if one hour is left before the deadline?

SCORING

Data truthfulness — 30% · silent failures — 20% · performance, UX and accessibility — 15% · live discussion — 35%. We're not just checking whether a metric improved. We want an engineer who questions assumptions, checks evidence, and understands what the system is actually doing.

Good luck. Be precise, honest, and curious.